



Department of Software Engineering

Faculty of Computing & Information Technology

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TO WHOM IT MAY CONCERN

This is to certify that Mr. Kashif Mahmood has been working as a Research Assistant under my supervision in the Department of Software Engineering, University of Sargodha, since September 2024. I have supervised his research through regular meetings and discussions and have been directly involved in the design and review of experiments during this period.

A major independent contribution was the conception, planning, and development of a peanut image dataset, carried out largely independently, including sourcing four Pakistani peanut varieties verified by the Barani Research Institute, Chakwal, and designing and executing an image-acquisition protocol spanning five backgrounds, four lighting conditions, and five quantity levels, resulting in 8,944 images. He identified a processing bottleneck and independently developed cropping software to address it, and personally collected morphometric measurements, including length, width, weight, seeds per pod, and shell thickness, using a digital caliper and precision weighing scale. He also developed comprehensive metadata and reproducibility documentation, including a CITATION.cff file, data card, license, and README. The dataset is now being used for baseline classification experiments. In parallel, he is engaged in a separate line of research in explainable artificial intelligence (XAI).

During this research period, Kashif has served as lead author on two manuscripts arising from his research work under my supervision. The first investigates deep-learning-based peanut disease detection using transfer learning and is under consideration at Scientific Reports. He was primarily responsible for the experimental work, evaluating EfficientNet-B0/B4, ConvNeXt-Base, and a ResNet-50 baseline in collaboration with researchers at the Islamic University of Madinah. The second, currently under revision at Scientific Reports, originated from a research question he independently conceived on chest X-ray classification under scarce annotations and scanner variability. He implemented four baseline architectures and self-supervised pretraining using SimCLR with ResNet-50 and MAE with ViT-S/16, evaluated across three random seeds and three annotation budgets, followed by fine-tuning and cross-validation across datasets, in partnership with researchers at Princess Nourah bint Abdulrahman University. He is currently managing the manuscript revision independently, with my oversight. He also contributed to a systematic review of visual computing in peanut agriculture (USJICT, under consideration), taking a leading role in its writing.

Both the dataset development and the self-supervised learning study originated from research questions proposed by Kashif rather than assigned tasks, reflecting a substantial degree of research independence. Should you require any further information, please feel free to contact me.

Supporting materials: Dataset: [Zenodo DOI](#) · Research Papers: [Peanut Disease Detection](#), [Chest X-Ray Classification](#), [Peanut Agriculture Systematic Review](#) · Software Code: [GitHub](#)

Best regards,

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